

INTERNSHIP MONTHLY REPORT

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Details of the internship schedule:

Key date	Key learning activities
Week 5 (4/8 - 8/8)	Learn to simulate a UAV on Gazebo simulation
Week 6 (11/8 - 15/8)	Apply autonomous flight mode for UAV in simulation, like onboard, offboard mission and auto.
Week 7 (18/8-22/8)	Deploy offboard flight on real hardware through Raspberry Pi
Week 8 (25/8 - 29/8)	Test flight with real S500 holybro using offboard and testing the camera and image processing

1. What skills have you learned in the past 4 weeks? Please describe your progress using the suggestions below:

I have been able to complete the assigned tasks according to the planned schedule each week, from simulation setup to real UAV testing. I consistently checked my work against documentation and testing results to ensure accuracy, and I maintained efficiency by troubleshooting errors early and seeking guidance when needed.

I collaborated effectively with my teammates by sharing progress updates, dividing tasks based on strengths, and supporting each other during problem-solving. I kept my work organized by setting weekly goals, documenting procedures, and following a step-by-step approach to ensure smooth transitions from simulation to hardware testing.

I took initiative by researching additional resources when encountering technical issues, such as communication errors between the Raspberry Pi and the flight controller. I also experimented with different configurations in Gazebo to better understand UAV responses. This proactive approach helped solve problems efficiently without delaying the team's progress.

I recognize the need to further improve my coding and debugging skills (especially for UAV software integration), as well as deepen my understanding of computer vision algorithms for image processing tasks. Additionally, I want to strengthen my PID tuning and control theory knowledge to achieve more stable and accurate UAV performance.

2. Are there any problems or delays with the internship where you need support?

Everything is going well .

With respect to the final report, with questions on professional responsibilities and on consideration of standards and laws:

During this internship, I ensured that my work was conducted responsibly and ethically. I took care to follow proper safety procedures while handling UAV hardware, particularly during test flights, to prevent accidents or damage. I also documented my work clearly so that teammates and future users could replicate or continue the project without ambiguity. Professional responsibility also meant respecting deadlines, collaborating effectively with my peers, and maintaining honesty in reporting both successful and unsuccessful test outcomes.